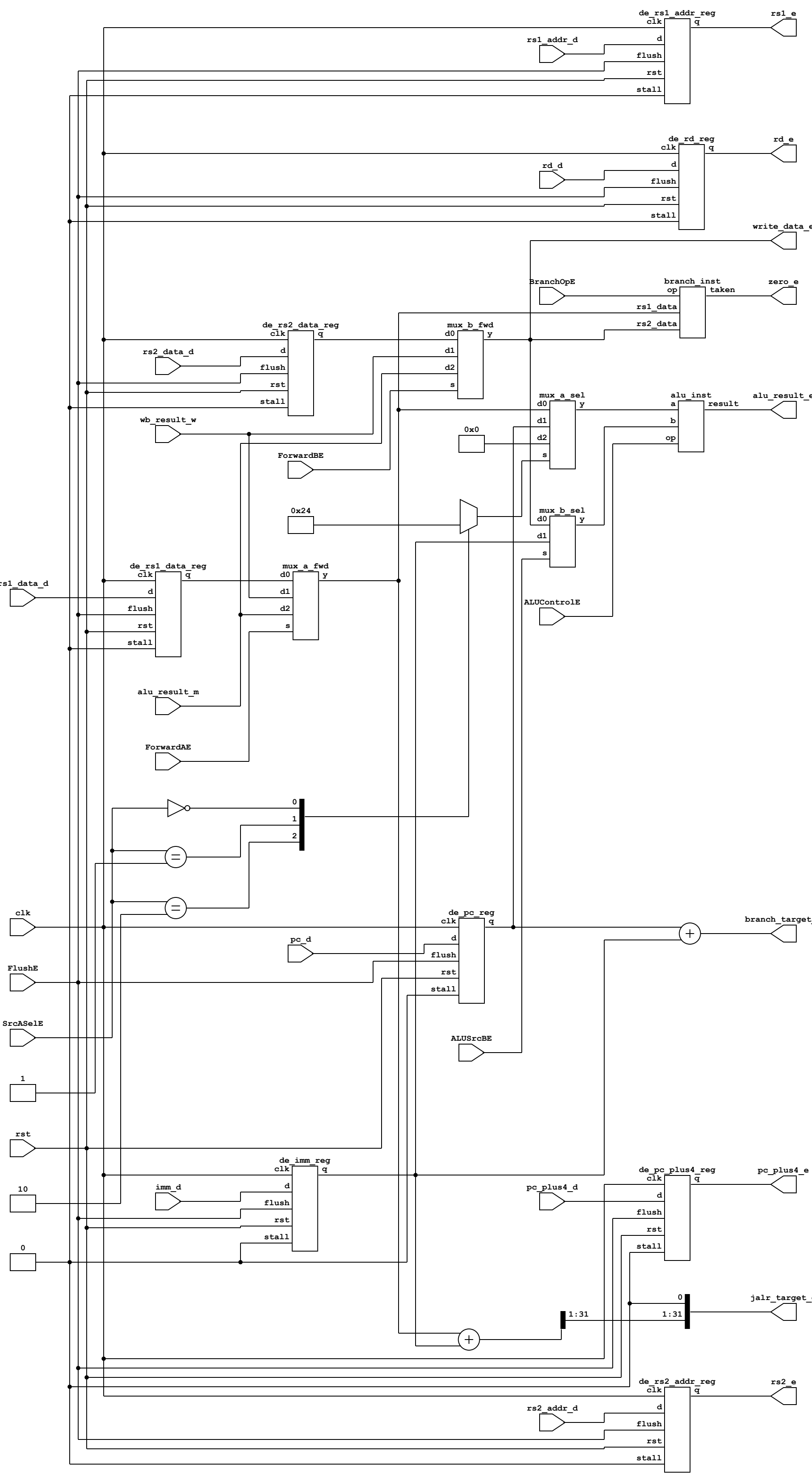
de_rs1_addr_reg
clk
q
rs1_e
rs1_addr_d
d
flush
rst
stall
de_rd_reg
clk
q
rd_e
rd_d
d
flush
rst
stall
write_data_e
BranchOpE
branch_inst
op
taken
zero_e
rs1_data
rs2_data
de_rs2_data_reg
clk
q
mux_b_fwd
d0
y
rs2_data_d
d
d1
flush
d2
rst
s
stall
mux_a_sel
d0
y
alu_inst
a
result
alu_result_e
wb_result_w
d1
b
0x0
d2
op
s
ForwardBE
0x24
mux_b_sel
d0
y
d1
s
de_rs1_data_reg
clk
q
mux_a_fwd
d0
y
rs1_data_d
d
d1
flush
d2
rst
s
stall
ALUControlE
alu_result_m
ForwardAE
0
1
2
clk
de_pc_reg
clk
q
branch_target_e
pc_d
d
flush
FlushE
rst
stall
SrcASelE
ALUSrcBE
1
rst
de_imm_reg
clk
q
10
imm_d
d
pc_plus4_d
de_pc_plus4_reg
clk
q
pc_plus4_e
flush
rst
0
stall
0
jalr_target_e
1:31
1:31
de_rs2_addr_reg
clk
q
rs2_e
rs2_addr_d
d
flush
rst
stall